Chapter 7

FINANCIAL MATHEMATICS

Learning objectives

After studying this chapter, the students will be able to understand

- Annuities - types of annuities
- Future and present values of annuities
- The concept of gain or loss in the sale and purchase of a stock.
- Brokerage in share transactions
- Effective rate of return

Introduction

In our day-to-day life we have lot of **money** transactions. In many transactions payment is made in single payment or in equal installments over a certain period of time. The amounts of these installments are determined in such a way that they compensate for their waiting time. In other cases, in order to meet future planned expenses, a regular saving may be done. (i.e) at regular time intervals, a certain amount may be kept aside, on which the person gains interest. In such cases the concept of annuity is used.

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7.1 Annuities

A sequence of equal payments made/ received at equal intervals of time is called **annuity**. The amount of regular payment of an annuity is called **periodic payment**. The time interval between two successive payments is called **payment interval** or **payment period**. Note that, the payment period may be annual, half yearly, quarterly, monthly (or) any fixed duration of time. The time interval between the first payment and the last payment of an annuity is called **term of an annuity**.

The sum of all payments made and interest earned on them at the end of the term of annuities is called future value of an annuity. The present or capital value of an annuity is the sum of the present values of all the payments of the annuity at the beginning of the annuity of purchase the payments due in future. Here we note that unless mentioned specifically, the payment means yearly payment.

7.1.1 Types of annuities

a) Based on the number of periods:

(i) Certain annuity: An annuity payable for a fixed number of years is called certain annuity.

Installments of payment for a plot of land, Bank security deposits, purchase of domestic durables are examples of certain annuity. Here the buyer knows the specified dates on which installments are to be made.

(ii) Annuity contingent: An annuity payable at regular interval of time till the happening of a specific event or the date

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